

Reinforcement Learning Gyms

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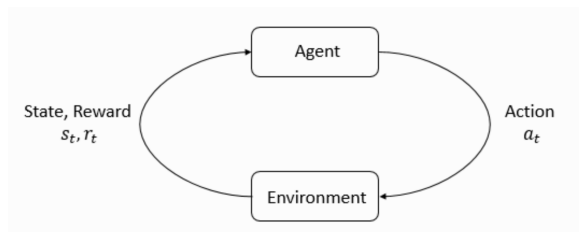
Overview

- Motivation: Why Reinforcement Learning Gyms
- Classical RL Gyms
 - ▶ Atari (discrete control)
 - ▶ MuJoCo (continuous control)
 - ▶ Isaac Lab (embodied & sim-to-real)
- RL for Large Language Models
 - ▶ RL from Human Feedback
 - ★ OpenRLHF
 - ★ LlamaFactory
 - ▶ RL from Verifiable Rewards
 - ★ TRL
 - ★ Verl
 - ★ Tinker
 - ★ SimpleGRPO

RL Gyms are Simulated Environments of the Real World

- Recall that Reinforcement Learning involves learning through trial and error.
- It requires collecting trajectories of states, actions, and rewards from the environment.
- AlphaGo Zero required **4.9M self-play games**^a.
- Such massive trial-and-error is inefficient and unsafe in the real world.
- RL researchers started with games...

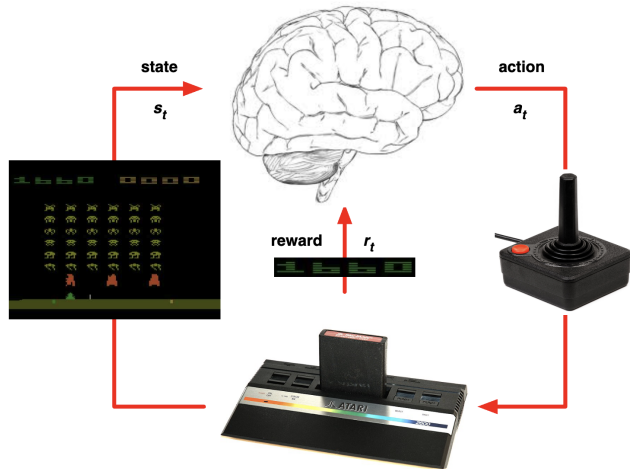
^asilver2017go.



- DeepMind aims to develop a general-purpose learning algorithm for Artificial General Intelligence.
- This requires a single algorithm that can solve diverse tasks at or above human level.
- Atari provides such a testbed: it contains 472 diverse video games, each requiring a different strategy.
- Deep Q-Network (DQN) was developed to solve Atari.
 - ▶ End-to-end learning: input raw video frames, output discrete control actions.
 - ▶ Achieved human-level or better performance in **29 of 49** benchmarked games¹.

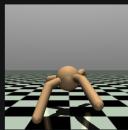
¹[mnih2015dqn](#).

RL Env: Atari

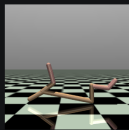


- Real-world control is continuous: you can move or rotate your arm by any arbitrary degree.
- In contrast, Atari's action space is discrete, e.g., left or right for Pong.
- **Mujoco** is proposed as a testbed for RL algorithms on continuous control problems.
- Trying to solve Mujoco led to algorithms like TRPO and PPO.
- PPO later becomes the de facto RL algorithm in many settings, including post-training LLMs.

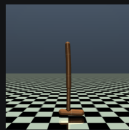
RL Env: Mujoco



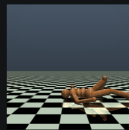
Ant



Half Cheetah



Hopper



Humanoid Standup



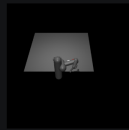
Humanoid



Inverted Double
Pendulum



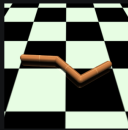
Inverted Pendulum



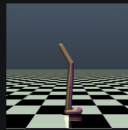
Pusher



Reacher



Swimmer



Walker2D

- Mujoco comes with simplified robots (HalfCheetah, 2D Walker).
- One can customize the .xml file to design their own robot in Mujoco.
- Yet, there is a notorious gap between simulation and reality in robotics.
- Issac Lab is designed to simulate real-world scenarios.



Starting Point to Understanding RL Code Implementation: CleanRL

- RL code has more complexity than supervised learning.
- Single File High Performance Implementation of popular RL algorithms
- For example, the `ppo_atari.py` only contains 340 lines of code.

CleanRL: Training Video Frames (Atari & MuJoCo)

- Train DQN on Breakout (Atari):

```
python cleanrl/dqn_atari.py --env-id BreakoutNoFrameskip-v4  
--capture-video
```

- Train PPO on HalfCheetah (MuJoCo):

```
MUJOOCO_GL=egl python cleanrl/ppo_continuous_action.py --env-id  
HalfCheetah-v4 --capture-video
```

- From LLM perspective, RL is the last stage of fine-tuning:
 - ▶ Stage 1: Pre-training with self-supervised learning on next token prediction on large corpus of text.
 - ▶ Stage 2: Supervised fine-tuning on human annotated responses.
 - ▶ Stage 3a: Reinforcement Learning from Human Feedback to further improve helpfulness, and reduce harmfulness.
 - ▶ Stage 3b: Reinforcement Fine-tuning with Verifiable Reward on Math and Coding to Incentivize Reasoning.

- From RL Perspective, natural language tasks fills a missing piece in existing RL Gyms
- Atari/Go/DOTA covers action space defined in human-designed games.
- Mujoco/IssacGYM covers continous control action space, robotic movement in the real-world.
- Missing out an essential action space for human: Language.

Reinforcement Learning from Human Feedback (RLHF)

- Stage 1: SFT.
- Stage 2: Reward Modeling. Typically follows the Bradley-Terry(BT) model to learn from human annotator's ranking responses.
- Stage 3: RL post-training. Further fine-tune the model with the guidance of previous learnt reward model; Can be online (e.g., PPO) or offline (e.g., DPO).

OpenRLHF

- Provides a stable RLHF implementation.
- Includes a built-in Reward Modeling pipeline (One key reason to use this library)
 - ▶ Originally designed for modeling human preference over responses
 - ▶ Now can be used for many other purposes, such as ...
 - ▶ Use `openrlhf/datasets/reward_dataset.py` with your customized instruction, accepted response, and rejected response to prepare the dataset
- Typically requires substantial GPU memory
 - ▶ Tip: use a remote reward model to reduce GPU usage

OpenRLHF / examples / scripts /	
<code>train_ppo_llama_ray_70b.sh</code>	support attn_implementation for GPT-OSS and flash_attn3
<code>train_ppo_llama_ray_hybrid_engine.sh</code>	support python -m and upgrade vllm
<code>train_ppo_llama_ray_hybrid_engine_dynamic_batch.sh</code>	support train_max_tokens_per_gpu and rollout_max_tokens_per
<code>train_ppo_llama_ray_ring.sh</code>	support attn_implementation for GPT-OSS and flash_attn3
<code>train_ppo_llama_ray_slurm.sh</code>	support attn_implementation for GPT-OSS and flash_attn3
<code>train_ppo_llama_ray_tensor_parallelism.sh</code>	update scripts
<code>train_ppo_llama_with_remote_rm.sh</code>	support attn_implementation for GPT-OSS and flash_attn3
<code>train_ppo_llama_with_reward_fn.sh</code>	support attn_implementation for GPT-OSS and flash_attn3
<code>train_prm_mistral.sh</code>	support attn_implementation for GPT-OSS and flash_attn3
<code>train_reinforce_baseline_llama_ray_agent_async.sh</code>	remove adam offload
<code>train_reinforce_baseline_llama_ray_async.sh</code>	remove adam offload
<code>train_reinforce_baseline_llama_ray_hybrid_engine.sh</code>	support attn_implementation for GPT-OSS and flash_attn3

OpenRLHF provides many example training scripts
(.sh)

Reinforcement Learning from Verifiable Reward

- RL from Human Feedback makes LLM a more helpful and less harmful conversational assistant.
- However, human preference is subjective and noisy for many real-world tasks.
- Use Rule-Based Reward as in Atari, Mujoco and Go,
- Solves hard problems like olympia math, coding and deeprersearch.
- Specifically, we a) enforce a format reward to require the LLM produce a chain-of-thought reasoning, and b) verify the correctness of the model generated answer with ground-truth answer.

Understanding RLVR Training: SimpleGRPO

- Task: 8 thousand Grade School Math problems (GSM8k).
- Format Reward:
- Correctness Reward:

RLVR Training Framework: SimpleGRPO

- Training Qwen-1.5B Model on GSM8k dataset
- curves.

List of RL-Fine-tuning Frameworks

- TRL: huggingface
- SimpleGRPO: for clear and ease of understanding
- Llama-Factory: with ktransformers, fine-tune trillion parameter model such as Kimi-k2.
- Verl: Multi-Turn Tool Calling support.
- Tinker: Stable and Robust RL fine-tuning.

A Natural Next Question

- Great RL Gyms inspire great algorithms
- To solve Atari, DQN is proposed.
- To solve Mujoco, PPO is proposed.
- DQN (or Rainbow DQN) and PPO are the most effective algorithms to solve Atari and Mujoco.
- People have since been applying PPO to other tasks (reasoning, real-world humanoid robot) with some success.
- Yet, they are old (from 2014, and 2017), and still far from artificial general intelligence (Most optimistic estimates are in future 5 - 10 years)
- What is missing in these RL Gyms?

Thanks!

Questions?

References I